USER MANUAL · REV 1.0

EP BOOSTER

Echo Plex Boost with MIDI · a faithful clone of the EP-style tape-preamp booster · handmade in Austria by Walter Craffonara

NOTE This pedal is a faithful clone of the EP-style tape-preamp booster — the preamp stage of the legendary Echoplex tape delay — expanded with a function I added: full MIDI control. The names “EP Booster” and “Echoplex” are used only to identify the original circuit this build recreates. CRAFFONARA FX is an independent builder and is not affiliated with, authorised by or endorsed by Xotic or Dunlop.

1 Overview & Concept

A discrete FET preamp with a low-impedance output, giving up to +20 dB of unadulterated boost that is multi-dimensional, with shimmering highs and lows — a faithful clone of the preamp stage of the legendary Echoplex tape delay.

Internal DIP switches choose the boost level and EQ setting. The pedal runs from 9 V up to 18 V for the headroom you want, and everything — effect state, boost and EQ — can also be controlled via MIDI over a TRS cable, including switching in time with the MIDI clock.

2 Controls & Connections

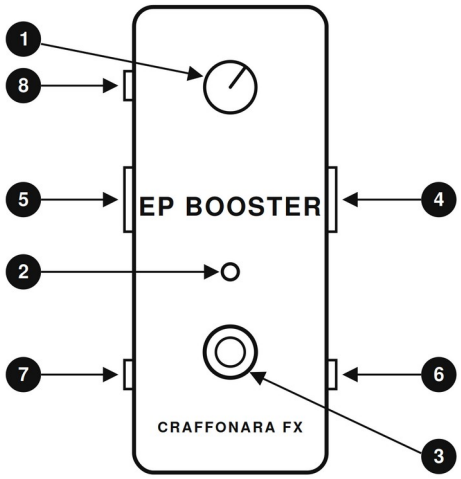


Fig. 1 — Panel and connections

1 LEVEL control sets the amount of boost, up to +20 dB.

2 STATUS LED off = bypass · green = effect on · orange = effect on with the +3 dB boost set via MIDI.

3 FOOTSWITCH silent footswitch, turns the effect on and off.

4 INPUT 6.35 mm mono jack.

5 OUTPUT 6.35 mm mono jack, low impedance.

6 MIDI IN TRS socket for incoming MIDI (Type A).

7 MIDI THRU TRS socket passing incoming MIDI on to the next device.

8 POWER 9–18 V DC, center-negative — 18 V gives more headroom.

3 Internal DIP Switches

The Boost (+3 dB) and Bright (EQ) parameters can be set with the internal DIP switches — or via MIDI.

Setting	DIP 1	DIP 2	Character
Default	ON	ON	+3 dB gain, flat EQ
Unity gain	OFF	ON	No extra gain, flat EQ
Vintage	OFF	OFF	No extra gain, vintage EQ

NOTE A parameter can only be changed via MIDI if its DIP switch is set to OFF.

DIP 1 controls the +3 dB boost, DIP 2 controls the Bright switch (ON = flat EQ, OFF = vintage voicing).

4 MIDI Reference

The pedal responds to MIDI Control Change (CC) messages on its MIDI channel (see the next section).

CC	Value	Command
10	00	Effect OFF
	01	Effect ON
	02	Toggle ON / OFF
	10–15	Toggle to the MIDI clock: 1/4 · 1/8 · triplet · 1/16 · dotted 1/8 · 1/32 notes
	16–17	Toggle to the MIDI clock: 1/2 notes · whole notes
	18–24	Toggle to the MIDI clock: every 2nd … every 8th whole note
30	00 / 01	Boost OFF / ON — DIP 1 must be OFF to control this via MIDI
40	00 / 01	Bright (EQ) OFF / ON — DIP 2 must be OFF to control this via MIDI

5 Setting the MIDI Channel

With the footswitch

1. Disconnect the pedal from the power supply.
2. Press and hold the FOOTSWITCH while connecting the power supply — the STATUS LED lights up and starts to flash, showing the pedal is in channel-set mode.
3. Press the FOOTSWITCH as many times as the channel number you want (e.g. twice for MIDI channel 2). The STATUS LED acknowledges with short flashes matching the chosen channel.
4. Press and hold the FOOTSWITCH until the STATUS LED goes dark and the pedal switches off completely.
5. Disconnect the supply. On the next power-up the pedal responds to the new channel.

NOTE To put the pedal into omni mode (it responds to every channel), skip step 3.

With MIDI commands

Send the following commands one after another, with no other command in between (<ch> is the desired channel):

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CC 06 <ch>    CC 09 18    CC 09 52    CC 09 08
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For omni mode send the same sequence with CC 06 00.

6 Power

Run the pedal on 9–18 V DC, center-negative. At 18 V the preamp has noticeably more headroom — a cleaner, more open boost. There is no battery compartment.

7 Specifications

Parameter	Value
Type	Clean boost — faithful clone of the EP-style tape-preamp circuit, with MIDI
Circuit	Discrete FET preamp, low-impedance output
Boost	Up to +20 dB; +3 dB stage and Bright EQ via DIP or MIDI
Switching	Silent footswitch; MIDI clock sync from 1/32 notes up to 8 bars
MIDI	IN + THRU (TRS, Type A), Control Change
Audio I/O	6.35 mm mono input and output
Power	9–18 V DC, center-negative (18 V = more headroom)
Enclosure	Brushed aluminium mini enclosure

NOTE Trademark notice: “EP Booster” is a product name of Xotic; “Echoplex” is a trademark of Dunlop Manufacturing, Inc. They are used here purely to describe the original circuit that this independent build recreates. CRAFFONARA FX is not affiliated with, authorised by or endorsed by Xotic or Dunlop.